

Goal

In this online exam, you must add an IP to the Aquila SoC to find the minimal and maximal components of a 1×8 integer vectors. The IP has eleven 32-bit registers that can be accessed by the processor through a memory-mapped I/O interface. A program ELF file (see Figure 1) will be given to you to test the IP. The program is compiled for DRAM so you should use the workspace of HW#3 as the starting point of your design and use circuit simulation to test the IP.

Exam Regulations

- [1] You are not allowed to use your mobile phone, browse the internet for any references, and talk to any other classmates. If you have any questions, please raise your hand and the TA will talk to you.
- [2] You are allowed **2 hours** for this online exam.
- [3] You can download code or the Power Point slides from E3 to the test computer between 10:10 am ~ 10:25 am. Any downloading behavior outside this period will be regarded as cheating.
- [4] After your circuit is completed and verified, please upload all your **modified** Verilog files to E3. You have to leave the computer classroom immediately afterwards.

Exam Descriptions

Your IP module, `find_min_max`, should have eleven 32-bit registers: `A[]`, `min`, `max`, and `trig`. All registers are set to zeros upon system reset. The functions of these registers are as follows:

1. `reg [31:0] A[0:7];` is a 1×8 vector register that the CPU can use to pass the “signed” 32-bit integer components of the input vector `A[]` to the IP. These registers are readable/writable.
2. `reg [31:0] min, max;` are 32-bit registers where the IP can store the minimal and maximal components of the input vector `A[]`. The CPU can read back these registers when the computation is done. These two registers are read-only for the CPU.
3. `reg [31:0] trig;` is a 32-bit signed register that when the value is set to non-zero by the CPU, the circuit `find_min_max` will begin to find the minimal and maximal components of `A[]` and saves the components in the `min` and `max` registers. As soon as the `min` and `max` registers contains the correct values, the IP should reset `trig` to 0. This register is readable/writable.

```
#include <stdio.h>

#define A      ((int volatile *) 0xC2000000)
#define min    ((int volatile *) 0xC2000020)
#define max    ((int volatile *) 0xC2000024)
#define trig   ((int volatile *) 0xC2000028)

int main(void)
{
    A[0] = 17, A[1] = -24, A[2] = 33, A[3] = 6;
    A[4] = 173, A[5] = -743, A[6] = 12, A[7] = 0;

    // Trigger the IP to find min/max values
    *trig = 1;
    while (*trig) /* busy loop */; // Waiting for IP to respond

    printf("The minimal value is: %d.\n", *min);
    printf("The maximal value is: %d.\n", *max);
}
```

Figure 1. The test program of the ELF file.

Note: TAs will use a different program to verify your code after the exam so you shall not hardcode the solution into to your IP.